

Bank Management System using JDBC

Code:

Connection Manager:

```
package com.mits.DAO;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;

public class ConnectionManager
{
    static Connection con=null;

    public static Connection getConnection() throws ClassNotFoundException,
SQLException
    {
        Class.forName("com.mysql.cj.jdbc.Driver");

        con=DriverManager.getConnection("jdbc:mysql://localhost:3306/BMA","root","root");

        con.setAutoCommit(false);

        return con;
    }

    public static void commit() throws SQLException
    {
        if(con!=null)
        {
            con.commit();
            con.close();
        }
    }

    public static void rollback() throws SQLException
```

```

        {
            if(con!=null)
            {
                con.rollback();
                con.close();
            }
        }
    }
}

```

Customer Service:

```

package com.mits.service;

import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.sql.Statement;
import com.mits.DAO.ConnectionManager;

public class CustomerService {

    public void insertData(int id,String name,double balance,String email,int
account_no,String password) throws SQLException, ClassNotFoundException
    {

        Connection connection=ConnectionManager.getConnection();

        PreparedStatement ps=connection.prepareStatement("insert into
customer values(?,?,?,?,?,?)");

        ps.setInt(1,id);

        ps.setString(2,name);

        ps.setDouble(3,balance);

        ps.setString(4,email);

        ps.setInt(5, account_no);

        ps.setString(6, password);
    }
}

```

```
        int status=ps.executeUpdate();
        if(status!=-1)
        {
            System.out.println("Transaction Successfull!");
        }
        else
        {
            System.out.println("transaction Failed!");
        }
    }
}
```

```
public void updateData(int id,String name,double balance,String email,int
account_no,String password) throws ClassNotFoundException, SQLException
```

```
{
    // TODO Auto-generated method stub
    Connection connection=ConnectionManager.getConnection();
    String query=("update customer set
name=?,balance=?,email=?,account_no=?,password=? where id=?");
    PreparedStatement ps=connection.prepareStatement(query);
    ps.setInt(6, id);
    ps.setString(1,name);
    ps.setDouble(2,balance);
    ps.setString(3,email);
    ps.setInt(4,account_no);
    ps.setString(5, password);
    int status=ps.executeUpdate();
    if(status>0)
    {
        System.out.println("Transaction Successfully..");
    }
}
```

```

        }
        else
        {
            System.out.println("Transaction failed..");
        }
    }

    public void deleteData(int id) throws ClassNotFoundException, SQLException
    {
        Connection connection=ConnectionManager.getConnection();
        PreparedStatement ps=connection.prepareStatement("delete from
customer where id=?");
        ps.setInt(1, id);
        int status=ps.executeUpdate();
        if(status!=-1)
        {
            System.out.println("Transaction Successfull.....");
        }
        else
        {
            System.out.println("Transaction Failed.....");
        }
    }

    public void readData() throws ClassNotFoundException, SQLException
    {
        Connection connection=ConnectionManager.getConnection();
        Statement st=connection.createStatement();
        ResultSet set=st.executeQuery("select *from customer");
    }

```

```

while(set.next())
{
    System.out.println("*****");
    System.out.println("Id: "+set.getInt(1));
    System.out.println("Name:"+set.getString(2));
    System.out.println("Balance:"+set.getDouble(3));
    System.out.println("Email:"+set.getString(4));
    System.out.println("Account No:"+set.getInt(5));
    System.out.println("Password:"+set.getString(6));
    System.out.println("*****");
}
}

```

```

public void displayData(String email, String password) throws
ClassNotFoundException, SQLException
{
    // TODO Auto-generated method stub

    Connection connection=ConnectionManager.getConnection();

    PreparedStatement ps=connection.prepareStatement("select balance
from customer where email=? and password=?");

    ps.setString(1, email);
    ps.setString(2, password);

    ResultSet rs=ps.executeQuery();

    if(rs.next())
    {
        System.out.println("Balance:"+rs.getDouble("balance"));
    }

    else
    {

```

```

        System.out.println("Invalid Email or password");
    }
}

}

```

Main function:

```

package com.mits.main;

import java.sql.SQLException;

import java.util.Scanner;

import com.mits.DAO.ConnectionManager;

import com.mits.service.CustomerService;

public class BankManagementSystem
{
    public static void main(String[] args) throws ClassNotFoundException,
    SQLException
    {
        Scanner sc=new Scanner(System.in);

        CustomerService cs=new CustomerService();

        while(true)
        {
            System.out.println("1.Insert data");

            System.out.println("2.Update data");

            System.out.println("3.Delete data");

            System.out.println("4.Read Data");

            System.out.println("5.Display Balance");

            System.out.println("6.Exit");

            int choice=sc.nextInt();

            switch(choice)

```

```

        {
            case 1->{
                System.out.println("=====Insert all
the data=====");
                System.out.println("Enter id:");
                int id=sc.nextInt();
                System.out.println("Enter name:");
                String name=sc.next();
                System.out.println("Enter balance:");
                double balance=sc.nextDouble();
                System.out.println("Enter Email:");
                String email=sc.next();
                System.out.println("Enter Account Number:");
                int account_no=sc.nextInt();
                System.out.println("Enter password:");
                String password=sc.next();
                cs.insertData(id,name,balance,email,account_no,password);
                ConnectionManager.commit();
            }
            case 2->
            {
                System.out.println("=====Update the
data=====");
                System.out.println("Enter name:");
                String name=sc.next();
                System.out.println("Enter Balance:");
                double balance=sc.nextDouble();
                System.out.println("Enter Email:");
                String email=sc.next();

```

```

        System.out.println("Enter Account number:");

        int account_no=sc.nextInt();

        System.out.println("Enter password:");

        String password=sc.next();

        System.out.println("Enter id");

        int id=sc.nextInt();

        cs.updateData(id,name,balance,email,account_no,password);

        ConnectionManager.commit();

    }

    case 3->

    {

        System.out.println("=====Delete
Data=====");

        System.out.println("Enter id:");

        int id=sc.nextInt();

        cs.deleteData(id);

        ConnectionManager.commit();

    }

    case 4->

    {

        System.out.println("=====Read the
data=====");

        cs.readData();

    }

    case 5->

    {

        System.out.println("=====Display all the
data=====");

```



```

        System.out.print("Enter Email:");

        String email=sc.next();

        System.out.print("Enter password:");

        String password=sc.next();

        cs.displayData(email,password);

    }

    case 6->

    {

        System.out.println("Thank you!");

        sc.close();

        System.exit(0);

    }

    default->System.out.println("Invalid choice");

}

}

}

```

Database

MySQL:

Create database:

Create database bma;

Use database:

Use bma;

Create a table:

Create table customer(id int, name varchar(50), balance double, email varchar(50), account_no int, password varchar(50);

Output: table created;

Output

Output from the code:

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

1

=====Insert all the data=====

Enter id:

101

Enter name:

Meghana

Enter balance:

10000

Enter Email:

megha@gmail.com

Enter Account Number:

98765

Enter password:

megha

Transaction Successfull!

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

1

=====Insert all the data=====

Enter id:

102

Enter name:

Hemanth

Enter balance:

20000

Enter Email:

hemu@gmail.com

Enter Account Number:

76543

Enter password:

hemu

Transaction Successfull!

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

1

=====Insert all the data=====

Enter id:

103

Enter name:

Manojj

Enter balance:

20000

Enter Email:

manoj@gmail.com

Enter Account Number:

12345

Enter password:

manoj

Transaction Successfull!

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

2

=====Update the data=====

Enter name:

Manoj

Enter Balance:

50000

Enter Email:

ManojNaik@gmail.com

Enter Account number:

123456

Enter password:

Manoj

Enter id

103

Transaction Successfully..

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

4

=====Read the data=====

Id: 101

Name:Meghana

Balance:10000.0

Email:megha@gmail.com

Account No:98765

Password:megha

Id: 102

Name:Hemanth

Balance:20000.0

Email:hemu@gmail.com

Account No:76543

Password:hemu

Id: 103

Name:Manoj

Balance:50000.0

Email:ManojNaik@gmail.com

Account No:123456

Password:Manoj

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

5

=====Display all the data=====

Enter Email:Manoj@gmail.com

Enter password:manoj

Invalid Email or password

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

5

=====Display all the data=====

Enter Email:hemu@gmail.com

Enter password:hemu

Balance:20000.0

- 1.Insert data
- 2.Update data
- 3.Delete data
- 4.Read Data
- 5.Display Balance
- 6.Exit

3

=====Delete Data=====

Enter id:

103

Transaction Successfull.....

- 1.Insert data
- 2.Update data
- 3.Delete data
- 4.Read Data
- 5.Display Balance
- 6.Exit

4

=====Read the data=====

Id: 101

Name:Meghana

Balance:10000.0

Email:megha@gmail.com

Account No:98765

Password:megha

Id: 102

Name:Hemanth

Balance:20000.0

Email:hemu@gmail.com

Account No:76543

Password:hemu

1.Insert data

2.Update data

3.Delete data

4.Read Data

5.Display Balance

6.Exit

6

Thank you!

[Output from the MySQL:](#)

```
mysql> select *from customer;
```

```
+----+-----+-----+-----+-----+-----+
| id | name  | balance | email      | account_no | password |
+----+-----+-----+-----+-----+-----+
| 101 | Meghana | 10000 | megha@gmail.com | 98765 | megha |
| 102 | Hemanth | 20000 | hemu@gmail.com | 76543 | hemu |
| 103 | Manojj | 20000 | manoj@gmail.com | 12345 | manoj |
+----+-----+-----+-----+-----+-----+
```

3 rows in set (0.00 sec)

```
mysql> select *from customer;
```



```

+----+-----+-----+-----+-----+-----+
| id | name  | balance | email          | account_no | password |
+----+-----+-----+-----+-----+-----+
| 101 | Meghana | 10000 | megha@gmail.com | 98765 | megha |
| 102 | Hemanth | 20000 | hemu@gmail.com  | 76543 | hemu  |
| 103 | Manoj   | 50000 | ManojNaik@gmail.com | 123456 | Manoj |
+----+-----+-----+-----+-----+-----+

```

3 rows in set (0.00 sec)

mysql> select *from customer;

```

+----+-----+-----+-----+-----+-----+
| id | name  | balance | email          | account_no | password |
+----+-----+-----+-----+-----+-----+
| 101 | Meghana | 10000 | megha@gmail.com | 98765 | megha |
| 102 | Hemanth | 20000 | hemu@gmail.com  | 76543 | hemu  |
+----+-----+-----+-----+-----+-----+

```

2 rows in set (0.00 sec)